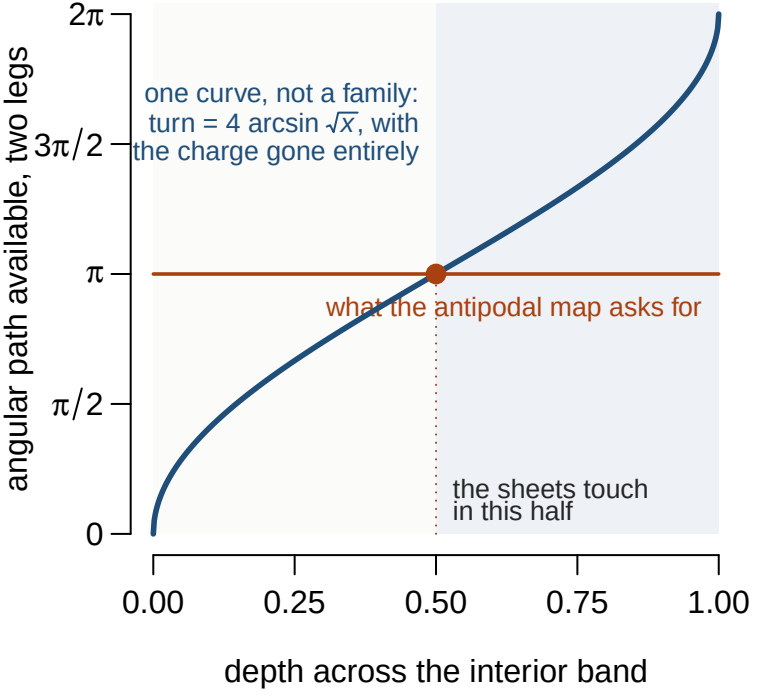
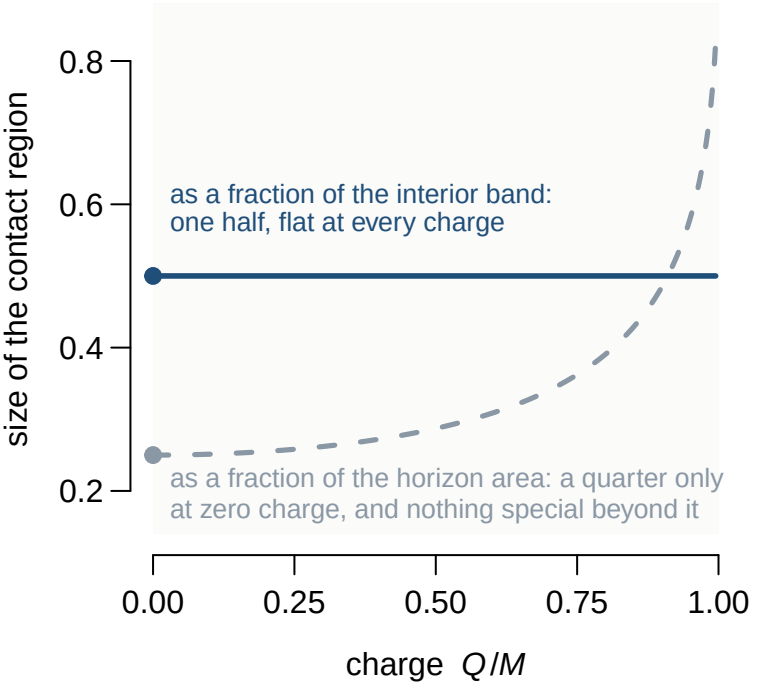
the curve crosses the line at the midpoint, at every charge

and the quarter−area coincidence does not survive it